

FRANCK DEBA

Quantitative Researcher | Financial Engineering & Data Science

@ debafranck@gmail.com

+33 (0) 6 05 77 47 51

Palaiseau, France

in linkedin.com/in/deba-franck

github.com/usernameFD

SUMMARY

Quantitative analyst with strong theoretical foundation in financial engineering and machine learning, currently pursuing M2MO master's degree. Proven expertise in derivatives pricing, risk modeling, and statistical analysis. Seeking quantitative researcher role to apply advanced analytical skills in financial markets.

EDUCATION

MSc in Quantitative Finance and Data Science (M2MO)	2025 – Present
Université Paris Cité	Paris, France
MSc in Data Science and Risk Management	2023 – 2025
ENSAI	Rennes, France
MSc in Financial Engineering	2024 – Present
WorldQuant University	Remote
MSc in Statistics and Economics	2021 – 2023
ISSEA - CEMAC	Yaoundé, Cameroon

TECHNICAL SKILLS

Quantitative Finance

- Stochastic Calculus & Ito's Lemma
- Derivatives Pricing Models
- Portfolio Optimization
- Risk Management (VaR, CVaR)

Programming

- Python: NumPy, Pandas, Scikit-learn
- C++/R/SQL/SAS
- Machine Learning
- Monte Carlo Methods

Financial Modeling

- Time Series Analysis
- Volatility Modeling
- Statistical Arbitrage
- Model Calibration

PROFESSIONAL EXPERIENCE

Quantitative Risk Analyst Intern	Apr 2025 – Sep 2025
Crédit Agricole	Paris, France
- Developed climate stress-testing framework assessing portfolio risk under NGFS scenarios - Engineered automated backtesting tool in Python (Dash) for Probability of Default models - Migrated statistical models from SAS to cloud-based Python platform, improving performance	
Quantitative Risk Analyst	Nov 2024 – Mar 2025
EY	Paris, France
- Developed PD scoring models using logistic regression for regulatory compliance (IRB) - Implemented machine learning models (XGBoost) for rating grade assignment - Challenged bootstrap methodologies for Model Uncertainty estimation	
Quantitative Modeling Intern	Jun 2024 – Aug 2024
Groupe VYV - Harmonie Mutuelle	Paris, France
- Applied unsupervised learning (K-Means) for insured population segmentation - Implemented survival analysis models (Cox PH) for risk exposure quantification - Developed GLMs for treatment cost forecasting by risk segment	

QUANTITATIVE PROJECTS

Stochastic Volatility Models & Exotic Derivatives | GitHub

- Calibrated Heston/Bates models using Fourier transform methods (Lewis, Carr-Madan)
- Priced exotic Asian options via Monte Carlo simulation with risk-neutral valuation
- Engineered CIR model calibration to Euribor term structure

Derivative Pricing & Portfolio Optimization | GitHub

- Built zero-coupon yield curves via bootstrapping methodology
- Implemented Black-Scholes & Hull-White models for derivatives pricing
- Developed portfolio optimization strategies tracking CAC40 index

Algorithmic Trading Strategy Backtest | GitHub

- Modeled stock returns using VAR model with FARIMA-GARCH volatility specification
- Implemented LSTM and Prophet models for multi-horizon price forecasting
- Developed Dash-based platform for strategy visualization and performance analysis

CERTIFICATIONS

- AMF Certification - Autorité des Marchés Financiers
- AI in Finance - World Bank
- Financial Markets - Yale University (Coursera)

LANGUAGES

- French: Native
- English: Professional Proficiency (TOEFL/TOEIC Certified)